

I. Single choice questions

1. A continuous-time signal is sampled at $f_s=400\text{Hz}$, yielding a sequence of 200 samples. After performing a 400-point DFT on this 200-sample sequence, the true physical frequency interval between two adjacent spectral lines in the frequency domain is (**C**)Hz.

- A. $\pi/100$ B. $\pi/200$ C. 1 D. 2

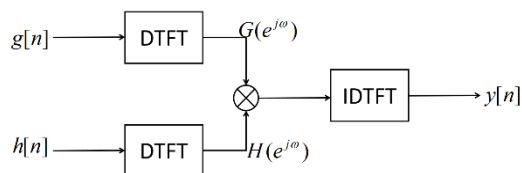
2. The discrete-time system $y[n] = x[n](u[n] - u[n-5])$ is (**A**).

- A. Linear time-varying B. Nonlinear time-varying
C. Linear time-invariant D. Nonlinear time-invariant

3. For a complex sequence $x[n]$ of length N , its N -point DFT is $X[k]$. Regarding the symmetry of $X[k]$, which of the following is **incorrect**? (**C**).

- A. $x^*[n] \xleftrightarrow{DFT} X^*[\langle -k \rangle_N]$ B. $x_{ca}[n] \xleftrightarrow{DFT} jX_{im}[k]$
C. $x_{im}[n] \xleftrightarrow{DFT} X_{ca}[k]$ D. $x_{re}[n] \xleftrightarrow{DFT} X_{cs}[k]$

4. As shown in the figure, which of the following correctly describes the relationship among $g[n]$, $h[n]$, and $y[n]$? (**A**).



- A. $y[n]$ is the linear convolution of $h[n]$ and $g[n]$
B. $y[n]$ is the circular convolution of $h[n]$ and $g[n]$
C. $y[n] = G(e^{j\omega})H(e^{j\omega})$ D. $y[n] = g[n]h[n]$

5. Given that the region of convergence for the Z-transform of a sequence is $|Z| > 0.5$, the sequence is (**B**).

- A. a finite-length sequence B. a right-sided sequence

C. a left-sided sequence

D. a two-sided sequence

6. Before sampling an analog signal, the signal must pass through an anti-aliasing filter.

This filter is (**A**).

A. a lowpass filter

B. a highpass filter

C. a multiband filter

D. a bandstop filter

7. Regarding the design of IIR filters using the bilinear transformation, which of the following statements is **correct**? (**D**).

A. The transformation used is a multi-valued mapping from the s-plane to the z-plane.

B. It is not recommended for designing bandpass and bandstop filters.

C. There is a linear relationship between digital frequency and analog frequency.

D. It maps a stable analog filter to a stable digital filter.

8. The inverse system of a causal linear-phase FIR filter is (**D**).

A. a causal stable system

B. a minimum-phase system

C. a maximum-phase system

D. not a causal stable system

9. When designing an IIR filter using the bilinear transformation, the relationship equation between the analog angular frequency Ω and the digital angular frequency ω is (**B**).

A. $\Omega = \frac{1}{T} \tan\left(\frac{\omega}{2}\right)$

B. $\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$

C. $\Omega = \frac{1}{T} \tan(\omega)$

D. $\Omega = \frac{2}{T} \tan(\omega)$

10. For several common analog prototype filters of the same order, the relationship between their transition bandwidths is (**D**).

A. Butterworth > Elliptic > Chebyshev

B. Chebyshev > Butterworth > Elliptic

C. Elliptic > Butterworth > Chebyshev

D. Butterworth > Chebyshev > Elliptic

II. Blank-fill questions

1. For a discrete-time LTI system describe by the difference equation

$y[n] - y[n-1] = \frac{1}{8}(x[n] - x[n-4])$, the unit impulse response of this LTI system is

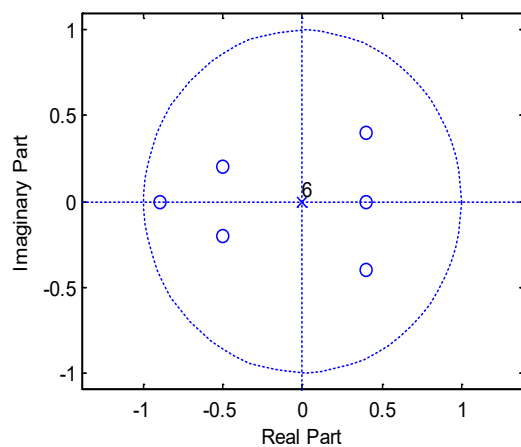
$$\{h[n]\} = \underline{\left\{\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right\}, n=0,1,2,3}$$

2. Given a complex sequence $v[n]=x[n]+jy[n]$ of length 4 with its 4-point DFT: $V[0]=-2+j6$, $V[1]=3+j5$, $V[2]=6+j4$, $V[3]=-1+j8$. Without computing the IDFT of $V[k]$, determine the 4-point DFTs of the real sequence $x[n]$ and $y[n]$:

$$X[k] = \underline{\{-2, 1-j3/2, 6, 1+j3/2\}} \quad Y[k] = \underline{\{6, 13/2-j2, 4, 13/2+j2\}}$$

3. IIR filter $y[n] = \sum_{k=0}^M p_k x[n-k] - \sum_{l=0}^N d_l y[n-l]$, which $M > N$, is realized by canonic structure, so the number of delay is M

4. A causal LTI system is as shown in the following pole-zero plot, so this system is a FIR (FIR or IIR) filter, and it is a minimum phase (linear phase, minimum phase or maximum phase) system.



5. The computational complexity of the complex addition operations in an N-point sequence FFT is $N \log_2 N$

6. A 8 points average filter is described by $y[n] = \frac{1}{8} \sum_{l=0}^7 x[n-l]$, please find out its group delay $\tau(w) = \underline{3.5}$

7. $x[n]$ is a causal sequence, its Z transform $X(z) = \frac{0.9 - z^{-1}}{1 - 0.9z^{-1}}$, so $x[1] = \underline{-0.19}$

8. When designing an IIR digital high-pass/band-pass/band-stop filter using the bilinear transform method, after obtaining the analog low-pass filter, one may either apply

analog spectral inversion or use Digital Spectrum Transformation to obtain the target IIR digital filter.

III. Calculations

1. Given a signal $x(t) = \cos\left(\frac{3\pi}{2}t\right) + \cos\left(\frac{39\pi}{8}t\right) + \sin(3\pi t)$

sampled at $f_s = 3\text{Hz}$,

1) Write the discrete-time sequence $x[n]$. Determine whether this signal is periodic. If yes, find its period.

2) Can the continuous-time signal $x(t)$ be recovered from $x[n]$? Explain your reasoning.

3) Write out the 16-point DFT of $x[n], 0 \leq N \leq 15$

Answer:

$$x(n) = x_c(t)|_{t=nT} = \cos\left(\frac{3\pi}{2} \cdot \frac{1}{3}n\right) + \cos\left(\frac{39\pi}{8} \cdot \frac{1}{3}n\right) + \sin\left(3\pi \cdot \frac{1}{3}n\right)$$

$$\begin{aligned} 1) &= \cos\left(\frac{\pi}{2}n\right) + \cos\left(\frac{13\pi}{8}n\right) + \sin(\pi n) \\ &= \cos\left(\frac{\pi}{2}n\right) + \cos\left(\frac{3\pi}{8}n\right) \end{aligned}$$

$$N = 16$$

2) Aliasing occurs, making it impossible to reconstruct $x(t)$ from $x[n]$.

$$\begin{aligned} 3) \quad DFT\{x(n)\} &= DFT\left\{\cos\left(\frac{\pi}{2}n\right) + \cos\left(\frac{3\pi}{8}n\right)\right\} \\ &= \begin{cases} 8, r=3,4,12,13 \\ 0, \text{其他} \end{cases} \end{aligned}$$

2. The autocorrelation sequence $c[n]$ of $x[n]$ is defined as

$$c[n] = \sum_{k=-\infty}^{+\infty} x(k)x(k-n)$$

Given that the Z-transform of $x[n]$ is $X(z)$, express the Z-transform of $c[n]$ in terms of $X(z)$.

Answer: $c[n] = x[-k] * x[k]$

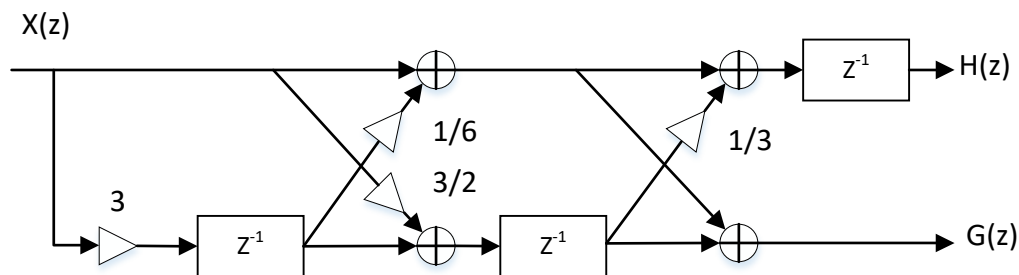
$$C(z) = X(z^{-1})X(z)$$

ROC: $(\text{RocX}, 1/\text{RocX})$ or including $(1/\text{RocX}, \text{RocX})$

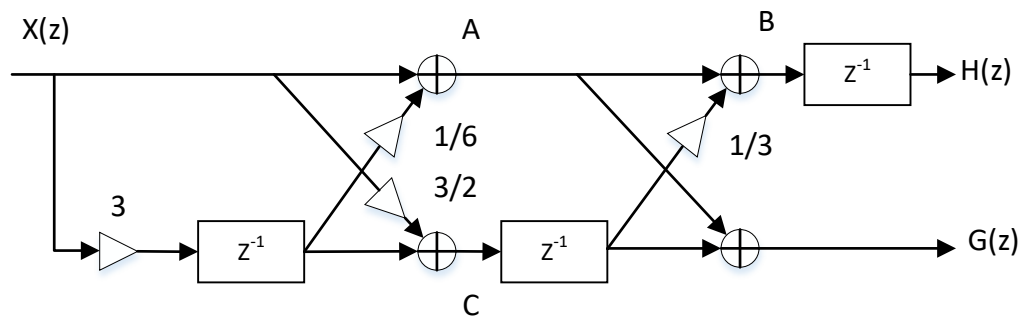
3. A filter has the following structure.

1) Determine its $H(z)$ and $G(z)$.

2) For $H(z)$, determine whether it is a canonical structure.



Answer:



$$A(z) = X(z) + 3 * \frac{1}{6} * X(z)z^{-1}$$

$$C(z) = \frac{3}{2}X(z) + 3 * X(z)z^{-1}$$

$$B(z) = A(z) + \frac{1}{3} * C(z)z^{-1}$$

$$G(z) = A(z) + C(z)z^{-1}$$

$$H(z) = B(z)z^{-1}$$

Then

$$1) H(z) = z^{-1} + z^{-2} + z^{-3}$$

$$G(z) = 1 + 2z^{-1} + 3z^{-2}$$

2) The order of $H(z)$ is 3, It's canonic

4. Find the system functions $H_1(z)$, $H_2(z)$, $H_3(z)$ and $H_4(z)$

for the four types of real-coefficient linear-phase FIR filters with the lowest order that has a zero at $z_1 = 0.3 + j0.4$.

Answer: According $|z_1| = 0.5$,

$$z_2 = z_1^* = 0.3 - j0.4, \quad z_3 = 1/z_1 = 1.2 - j1.6, \quad z_4 = z_3^* = 1.2 + j1.6$$

$$1) H_1(z) = \prod_{i=1}^4 (1 - z_i z^{-1}) = 1 - 3z^{-1} + 5.69z^{-2} - 3z^{-3} + z^{-4}$$

$$2) z_5 = -1$$

$$H_2(z) = (1 + z^{-1})H_1(z) = 1 - 2z^{-1} + 2.69z^{-2} + 2.69z^{-3} - 2z^{-4} + z^{-5}$$

$$3) z_5 = -1, \quad z_6 = 1$$

$$\begin{aligned} H_3(z) &= (1 + z^{-1})(1 - z^{-1})H_1(z) \\ &= 1 - 3z^{-1} + 4.69z^{-2} - 4.69z^{-4} + 3z^{-5} - z^{-6} \end{aligned}$$

$$4) z_5 = 1$$

$$H_4(z) = (1 - z^{-1})H_1(z) = 1 - 4z^{-1} + 8.69z^{-2} - 8.69z^{-3} + 4z^{-4} - z^{-5}$$

5. The signal $y[n]$ consists of an original signal $x[n]$ and two echoes:

$$y[n] = x[n] + 0.5x[n - n_d] + 0.25x[n - 2n_d]$$

Design a filter to recover $x[n]$ from $y[n]$, and determine its stability.

Answer: $Y(z) = (1 + 0.5z^{-n_d} + 0.25z^{-2n_d})X(z)$

So

$$\begin{aligned} G(z) &= \frac{1}{1 + 0.5z^{-n_d} + 0.25z^{-2n_d}} \\ G(z) &= F(z^{n_d}), \quad F(z) = \frac{1}{1 + 0.5z^{-1} + 0.25z^{-2}} \end{aligned}$$

The poles of $F(z)$ are $z = -0.25(1 \pm j\sqrt{3})$

So the poles of $G(z)$ are $r' = (0.5)^{-n_d}$ it's stability.

6. The specification for a multiband filter are given as:

$$\omega_{s1} = 0.08\pi, \quad \omega_{p1} = 0.12\pi, \quad \omega_{p2} = 0.28\pi, \quad \omega_{s2} = 0.32\pi,$$

$$\omega_{s3} = 0.58\pi, \quad \omega_{p3} = 0.62\pi, \quad \omega_{p4} = 0.76\pi, \quad \omega_{s4} = 0.84\pi,$$

with all passband and stopband ripples constrained to be less than 0.01. Using the window method and selecting an appropriate window from the Appendix, design a linear-phase FIR filter with minimum length that satisfies these specifications.

Answer:

1)

$$\alpha_s = \alpha_p = -20 \log_{10}(0.01) = 40 \text{ dB}$$

2) Choose Hann window

$$w[n] = \frac{1}{2} \left[1 + \cos\left(\frac{2\pi n}{2M+1}\right) \right], \quad -M \leq n \leq M$$

3)

$$\Delta\omega_1 = \Delta\omega_2 = \Delta\omega_3 = 0.04\pi, \quad \Delta\omega_4 = 0.06\pi$$

$$3.11\pi/M \leq 0.04\pi \text{ then } M \geq 77.75, \quad M=78$$

4)

$$\omega_c = (\omega_s + \omega_p) / 2$$

$$\omega_{c1} = 0.1\pi, \quad \omega_{c2} = 0.3\pi$$

$$\omega_{c1} = 0.6\pi, \quad \omega_{c2} = 0.8\pi$$

$$h[n] = \frac{\sin(0.3\pi n)}{\pi n} - \frac{\sin(0.1\pi n)}{\pi n} + \frac{\sin(0.8\pi n)}{\pi n} - \frac{\sin(0.6\pi n)}{\pi n}$$

5)

$$h_d[n] = h[n]w[n]$$

$$h_d[n] = h_d[n-M]w[n-M], \quad 0 \leq n \leq 2M$$

Appendix:

Table1 Some fixed window functions

Rectangular:	$w[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$
hann:	$w[n] = \frac{1}{2} [1 + \cos(\frac{2\pi n}{2M+1})], -M \leq n \leq M$
hamming:	$w[n] = 0.54 + 0.46 \cos(\frac{2\pi n}{2M+1}), -M \leq n \leq M$
blackman:	$w[n] = 0.42 + 0.5 \cos(\frac{2\pi n}{2M+1}) + 0.08 \cos(\frac{4\pi n}{2M+1}), -M \leq n \leq M$

Table 2 Properties of some fixed window functions

Type of window	Main lobe width Δ_{ML}	Relative Sidelobe Level $A_{s\ell}$	Minimum Stopband Attenuation	Transition Bandwidth $\Delta\omega$
Rectangular	$4\pi/(2M+1)$	13.3dB	20.9dB	$0.92\pi/M$
Hann	$8\pi/(2M+1)$	31.5dB	43.9dB	$3.11\pi/M$
Hamming	$8\pi/(2M+1)$	42.7dB	54.5dB	$3.32\pi/M$
Blackman	$12\pi/(2M+1)$	58.1dB	75.3dB	$5.56\pi/M$